

Transposition Ciphers

Rail Fence Cipher

Mechanism

The Rail Fence Cipher is a transposition cipher that writes the plaintext in a zigzag pattern across multiple "rails" (rows) and then reads it off row by row.

How It Works

1. Choose the number of rails.
2. Write the plaintext letters diagonally across the rails.
3. When reaching the top or bottom rail, change direction.
4. Read the text row by row to get the ciphertext.

Example

Plaintext: WE ARE DISCOVERED. RUN AT ONCE

Rails: 3

Write the text in a zigzag pattern:

```
W . . . E . . . C . . . R . . . U . . . O . . .  
. E . R . D . S . O . E . E . R . N . T . N . E  
. . A . . . I . . . V . . . D . . . A . . . C .
```

Ciphertext: WECRUO ERDSOEERNTNE AIVDAC

Scytale Cipher

Mechanism

The Scytale Cipher involves wrapping a strip of paper around a rod of a specific diameter. The plaintext is written across the rod, and the ciphertext is obtained by unwrapping it.

How It Works

1. Determine the rod's diameter (this decides how many characters fit in one rotation).
2. Write the plaintext along the length of the rod.
3. Unwrap the paper to get the ciphertext.

Example

Plaintext: I am hurt very badly help

Rod Diameter: 4

Write the text in rows:

```
-----  
  |   |   |   |   |   |   |  
  | I | a | m | h | u |   |  
--| r | t | v | e | r |--|  
  | y | b | a | d | l |  
  | y | h | e | l | p |  
  |   |   |   |   |   |  
-----
```

Ciphertext: Iryyatbhmvaehedlurlp

Route Cipher

Mechanism

The Route Cipher involves writing the plaintext in a grid and reading it according to a specific route (e.g., spiral, zigzag).

How It Works

1. Create a grid with the plaintext.
2. Follow the chosen route to extract the ciphertext.

Example

Plaintext: MEET ME AFTER MIDNIGHT

Grid: 4x5

```
M E E T M  
E A F T E  
R M I D N  
I G H T
```

Route: Spiral starting top-left

Ciphertext: MEETMEFTAFRMIDNGHT

Columnar Transposition

Mechanism

In Columnar Transposition, the plaintext is written in rows under column headers. The ciphertext is formed by reading the columns in a specified order.

How It Works

1. Choose a keyword.
2. Write the plaintext in rows under the keyword's letters.
3. Number the columns based on the alphabetical order of the keyword.
4. Read the columns in the numbered order.

Example

Plaintext: ATTACK AT DAWN

Keyword: ZEBRA

Z	E	B	R	A
A	T	T	A	C
K	A	T	D	A
W	N			

Order: 5, 3, 2, 4, 1

Ciphertext: CATWATTNDAKA

Double Transposition Cipher

Mechanism

The Double Transposition Cipher applies two successive transpositions to the plaintext, using two different keys.

How It Works

1. Write the plaintext in a grid using the first key.
2. Perform a columnar transposition using the first key.
3. Use the resulting text to perform another columnar transposition using the second key.

Example

Plaintext: DEFEND THE EAST WALL

First Key: ZEBRA

Second Key: ALPHA

First transposition:

Z	E	B	R	A
D	E	F	E	N
D	T	H	E	E
A	S	T	W	A
L	L			

Ciphertext after first transposition: NEAFHTETSLEEWDDAL

Second transposition (using ALPHA):

A	L	P	H	A
N	E	A	F	H
T	E	T	S	L
E	E	W	D	D
A	L			

Final Ciphertext: NTEAHLDFSDEEELATW

Disrupted Transposition Cipher

Mechanism

In the Disrupted Transposition Cipher, the plaintext is written in rows, but some rows are shifted ("disrupted") based on a predefined pattern.

How It Works

1. Write the plaintext in rows.
2. Shift certain rows by a specified number of columns.
3. Read the text column by column to create the ciphertext.

Example

Grid:

F	O	R	E	V	E	R	J	I	G	S	A	W
4	8	9	2	12	3	10	7	6	5	11	1	13
C	O	M	P	L	I	C	A	T	E	S	T	*
H	E	T	R	*	*	*	*	*	*	*	*	*
A	N	S	P	O	S	*	*	*	*	*	*	*
I	*	*	*	*	*	*	*	*	*	*	*	*
T	I	O	N	P	A	T	T	E	R	*	*	*
N	L	I	K	E	A	C	O	M	*	*	*	*
B								*	*	*	*	*

The columns are then taken off as per regular columnar transposition: TPRPN, KISAA, CHAIT, NBERT, EMATO, etc.